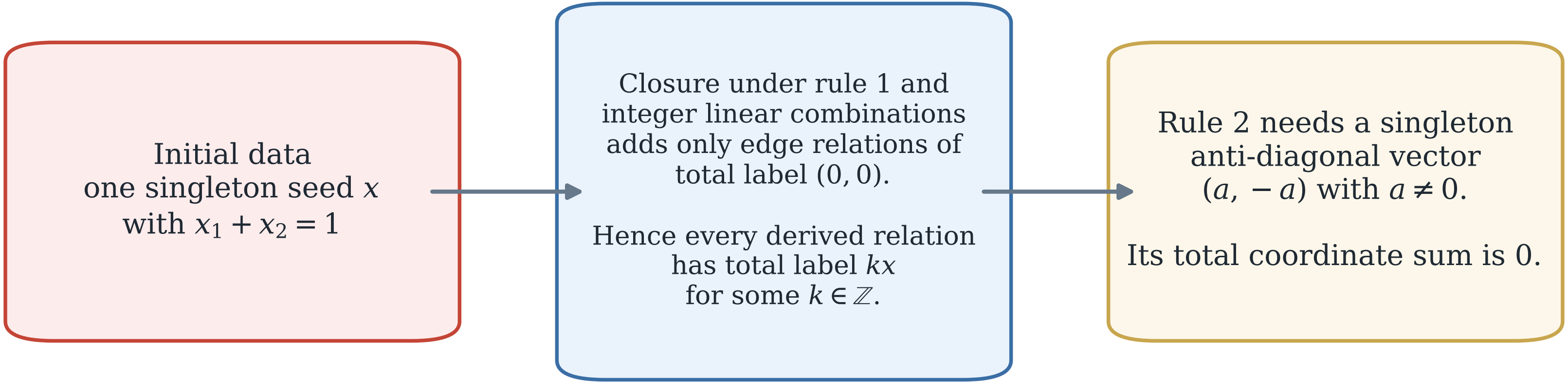
Coordinate-sum invariant



If $kx = (a, -a)$ then $k(x_1 + x_2) = 0$. Since $x_1 + x_2 = 1$, we obtain $k = 0$, hence $(a, -a) = (0, 0)$, contradicting rule 2.